UNIVERSITY OF CAPE COAST
DEPARTMENT OF MATHEMATICS
MAT101: ALGEBRA AND TRIGONOMETRY
TUTORIAL SET 1

Attempt all questions.

1. Given that A, B and C are subsets of U. Show that:

   (a) i. $A \cap B = B \cap A$
   ii. $A \cup B = B \cup A$

   (b) $A \cup (A \cap B) = A \cap (A \cup B)$

2. Given $A = \{1, 2, 3, \cdots, 10\}$ and $B = \{2, 4, 6, \cdots\}$. Find
   a. $A \cap B$. b. $A \cup B$
   c. $(A \cap B)'$ d. $(A \cup B)'$
   e. $A' \cap B'$ f. $A' \cup B'$

3. Given that A, B and C are sets, prove the following:

   (a) $(A \cup B) \cup C = A \cup (B \cup C)$
   (b) $(A \cap B) \cap C = A \cap (B \cap C)$
   (c) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (d) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
   (e) $(A \cap B)' = A' \cup B'$
   (f) $(A \cup B)' = A' \cap B'$
   (g) $A \cap (A \cap B')' = A \cap B$
   (h) $A \cap (B \cup C)' = (A \cap B') \cap (A \cap C')$
   (i) $A \cap (B \cap C)' = (A \cap B') \cup (A \cap C')$

4. Let A and B be any two sets, prove whether or not

   (a) $(A - B) \cap (A - C) = A - (B \cup C)$
   (b) $(A - B) \cup (A - C) = A - (B \cap C)$

5. Prove that $A \cup (A \cap B) = A \cap (A \cup B) = A$

6. Verify that the following results are true

   (a) $(A \cup B \cup C)' = A' \cap B' \cap C'$
   (b) $(A \cap B \cap C)' = A' \cup B' \cup C'$

   In each case illustrate the results using Venn diagrams

7. (a) Prove that for the sets A and B, A x $B = B$ x A if and only if $A = B$ or at least one is empty.
   (b) Prove that $(A$ x $B)$ x $C = A$ x $(B$ x $C)$ if and only if at least one is empty.

8. (a) List all the members of A x B if
   $A =$ { first four natural numbers }
   $B =$ { integers between -2 and 2 }.
   (b) If $A =$ {-5,3}, $B =$ {2,4,6} and $C =$ {0,1}. Find A x B x C.

9. Let $A = \{1 \leq x \leq 3\}$, $B = \{2 \leq x \leq 3\}$. Sketch the graphs $A \times B$ and $B \times A$ in the plane. Describe your sketch in each case.

10. Let $A = \{1 \leqslant x \leqslant 2\}$, $B = \{0 \leqslant x \leqslant 1\}$ and $C = \{-1 \leqslant x \leqslant 2\}$. Sketch in the xyz space the graph of $A \times B \times C$. Describe your sketch.

11. Given that $A = \{3, 5\}$ and $B = \{2, 4, 6\}$, find

    (a) $A \times B$

    (b) $B \times A$

    (c) What conclusion can you draw from (a) and (b).

12. If $n(A - B) = 18, n(A \cup B) = 70$ and $n(A \cap B) = 25$, then find $n(B)$

13. As part of St. Ann's College custom, the college awarded medals in different categories. 35 medals in Music, 13 Medals in Dance and 18 medals in Poetry. If these medals went to a total of 46 students and only 5 got medals in all the three categories, how many received medals in exactly two of the categories?

14. In a group of students, 25 study Computer, 28 study Health, 20 study Mathematics, 9 study Computer only, 12 study Health only, 8 study Computer and Health only and 5 study Health and Mathematics only.

    (a) Draw a Venn diagram to illustrate the above information.

    (b) How many students are there altogether?

    (c) Find how many students study all the subjects.

15. A survey of the characteristics of 100 small businesses that have failed revealed that 95 of them either were overcapitalized, had inexperienced management, or had a poor location. Four of the businesses had all three of these characteristics. Forty businesses were overcapitalized but had experienced management and good location. Fifteen businesses had inexperienced management but sufficient capitalization and good location. Seven were overcapitalized and had poor location. Ten had inexperienced management and poor location. How many of the businesses had poor location? Which of the three characteristics was most prevalent in the failed businesses?

16. A study in the Department of Languages and Linguistics at the University of Cape Coast shows that of the 400 students in the department, 200 take French(F), 300 take English(E), and 50 take Twi(T). The study shows that 150 take both French and English, 30 take both French and Twi, and 20 take both English and Twi. Five students take all the three Languages. Use a Venn diagram to find the number of people in the sets given in (i) to (v).

    i. $T \cap (F \cup E)$

    ii. $(T \cup F \cup E)'$

    iii. $T \cup E \cup F'$

    iv. T'

    v. $F \cap E' \cap T'$

17. The cornea, the pupil and the retina have shown to be the most affected part of the eye of any glaucoma patient. A study at ANI eye care has revealed that out of some selected glaucoma patients, 19 were defected in the cornea, 27 in the pupil and 21 in the retina. 23 of the patients were encountered with at least two defections whilst 17 were encountered with exactly two of the defections. How many glaucoma patients had defections in the cornea, pupil and retina? Hence, find the number of glaucoma patients used in the study if all the patients had at least one of the defections.

18. Given the following functions $f(x) = 3x^3 - 2$, $g(x) = \sqrt{x+1}$ and $h(x) = \dfrac{2}{x^2}$

    (a) Which of the functions is/are even, odd or neither?

    (b) Find the domain and inverse of the functions

    (c) Determine $f\ o\ g(x)$, $h\ o\ g(x)$, $h\ o\ h(x)$ and hence evaluate $g\ o\ h\ o\ f(x)$ at $x = \dfrac{1}{2}$.

19. If $f(x) = 2x^3 + x - 3$ and $g(x) = x - 1$, evaluate $f(x).g(x) - [f(x) + g(x)]$.

20. Given that $f(x) = 2x + 1$, find $g(x)$ if $g(x) = 2[f(x)]^2 - 1$.

21. Evaluate

$$f(x) = \begin{cases} 5x - 2, & \text{x} < 6 \\ x^2, & \text{x} \geq 6 \end{cases}$$

    Find $f(-10)$, $f(12)$ and $f(0)$.

22. Find the domain and the inverse of the following functions

    (a) $f(x) = 6x^2 - 7x + 1$

    (b) $f(x) = (x-3)^3 + 1$

    (c) $\sqrt{4x^2 - y} = 2$

    (d) $f(x) = \dfrac{5 - 3x}{5x - 3}$

    (e) $\dfrac{2}{3\sqrt{x+5}} = 3y$

23. The lines $4x + 5y - 5 = 0$ and $2x + 4y - 3 = 0$ intersect at N. Find the equation of the line through N perpendicular to the line $x + 2y = 0$.

24. Graph the following functions showing the intercepts and the vertex where applicable

    (a) $f(x) = x^2 + 2x - 2$

    (b) $y = -\sqrt{x+3}$

    (c) $y = |x - 20| - 20$

    (d) $f(x) = \dfrac{x^3}{x^2 - 16}$

    (e) $f(x) = \dfrac{2}{1 - x^2}$

    (f) $f(x) = \dfrac{x}{x^2 - x^x - 2}$

25. Solve the following equations using the method of completing the square

    (a) $x^2 + 4x = 2$

    (b) $x^2 - 6x - 15 = 0$

    (c) $f(x) = 20x^2 + 32x - 3$

    (d) $(y-2)^2 = 4y - 12$

    (e) $\sqrt{x-4} + x = 6$

    (f) $\sqrt{6 - 2x} + x = 2(x + 15) - 9$

26. A function $f(x) = 4x^2 + kx + 9$. Find the values of k, If x-axis is a tangent to the curve.

27. Given that $x^2 + 2x + 3 - k(x^2 - 2x - 3) = 0$ always has real roots, find the range of values of k.

28. What are the values of h and k if $2x^3 - 10x^2 + 11x - 7 = (x-4)(2x^2 + hx + 3) + k$.

29. Given $z(x) = 6x^3 + bx^2 - 52x + 15$, $z(2) = 35$ and $z(-5) = 0$. Determine all the zeros of $z(x)$.

30. Factorize completely and sketch the graphs of

    (a) $x^4 + 6x^2 - 7 = 0$ (b) $x^3 - 4x^2 + 4x - 6 = 0$ (c) $4x^3 - x^2 + 16x - 4 = 0$

31. Given that $h(a) = 6a^3 + 11a^2 - 4a - 9$ and $a = \dfrac{2}{3}$ is a root, write $h(a)$ as $q(a) + \dfrac{r(a)}{d(a)}$.

32. If α and β are the roots of the equation $4x^2 + 7x + 3 = 0$, Find the value of $\dfrac{1}{\alpha^2} + \dfrac{1}{\beta^2}$.